Module 34

Application Design and Development/4: Python and PostgreSQL

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Module Recap

- Introduced the use of SQL from a programming language

Module Objectives

- To understand how to access PostgreSQL database from Python
- To understand Python Web Application with PostgresSQL

Module Outline

- Accessing PostgreSQL from Python
- Developing Web Application with Python

Working with PostgreSQL and Python

Python Modules for PostgreSQL

- Following Python modules that can be used to work with a PostgreSQL database server:
  - psycopg2
  - pg8000
  - py-postgresql
  - PyGreSQL
  - ocpgdb
  - bpgsql
  - SQLAlchemy (needs any of the above to be installed separately)

Source: `https://pynative.com/python-postgresql-tutorial/`

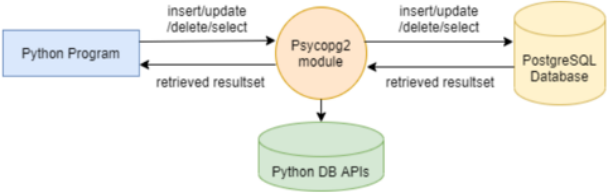
Package psycopg2

- **Advantages of psycopg2**
  - Most popular and stable module to work with PostgreSQL
  - Used in most of the Python and Postgres frameworks
  - An actively maintained package and supports Python 2.x and 3.x
  - Thread-safe and designed for heavily multi-threaded applications.
- **Installing Psycopg2 using pip command**
  - The following pip command installs psycopg2 on different operating systems including Windows, MacOS, Linux, and Unix:
  - `pip install psycopg2`
  - For installing specific version, the following command can be used:
  - `pip install psycopg2==2.8.6`

Source: `https://pynative.com/python-postgresql-tutorial/`

Steps to access PostgresSQL from Python

- **Steps to access PostgresSQL from Python using Psycopg**
  - a) Create connection
  - b) Create cursor
  - c) Execute the query
  - d) Commit/rollback
  - e) Close the cursor
  - f) Close the connection



Source: `https://pynative.com/python-postgresql-tutorial/`

Python psycopg2 Module APIs: connection objects

- `psycopg2.connect(database="mydb", user="myuser", password="mypass", host="127.0.0.1", port="5432")`
  - This API opens a connection to the PostgreSQL database. If database is opened successfully, it returns a connection object.
- `connection.close()`
  - This method closes the database connection.
- **Important psycopg2 module routines for managing cursor object:**
- `connection.cursor()`
  - This routine creates a cursor which will be used throughout the program.
- `cursor.close()`
  - This method closes the cursor.

Source: `https://www.tutorialspoint.com/postgresql/postgresql_python.htm`

Python psycopg2 Module APIs: insert, delete, update & stored procedures

- `cursor.execute(sql [, optional parameters])`
  - This routine executes an SQL statement. The SQL statement may be parameterized (i.e., placeholders instead of SQL literals). The psycopg2 module supports placeholder using %s sign. For example: `cursor.execute("insert into people values (%s, %s)", (who, age))`
- `cursor.executemany(sql, seq of parameters)`
  - This routine executes an SQL command against all parameter sequences or mappings found in the sequence SQL.
- `cursor.callproc(procname[, parameters])`
  - This routine executes a stored database procedure with the given name. The sequence of parameters must contain one entry for each argument that the procedure expects.
- `cursor.rowcount`
  - This is a read-only attribute which returns the total number of database rows that have been modified, inserted, or deleted by the last `execute()`.

Python psycopg2 Module APIs: select

- `cursor.fetchone()`
  - This method fetches the next row of a query result set, returning a single sequence, or None when no more data is available.
- `cursor.fetchmany([size=cursor.arraysize])`
  - This routine fetches the next set of rows of a query result, returning a list. An empty list is returned when no more rows are available. The method tries to fetch as many rows as indicated by the size parameter.
- `cursor.fetchall()`
  - This routine fetches all (remaining) rows of a query result, returning a list. An empty list is returned when no rows are available.

Python psycopg2 Module APIs: commit & rollback

- `connection.commit()`
  - This method commits the current transaction. If you do not call this method, anything you did since the last call to `commit()` is not visible to other database connections.
- `connection.rollback()`
  - This method rolls back any changes to the database since the last call to `commit()`.

Connect to a PostgreSQL Database Server

```
import psycopg2
def connectDb(dbname, usrname, pwd, address, portnum):
  conn = None
  try: # connect to the PostgreSQL database
    conn = psycopg2.connect(database = dbname, user = usrname, \
                            password = pwd, host = address, port = portnum)
    print ("Database connected successfully")
  except (Exception, psycopg2.DatabaseError) as error:
    print(error)
  finally:
    if conn is not None:
      conn.close() # close the connection

connectDb("mydb", "myuser", "mypass", "127.0.0.1", "5432") # function call
```

Steps to execute SQL commands

1. Use the `psycopg2.connect()` method with the required arguments to connect PostgresSQL. It would return an Connection object if the connection established successfully.
2. Create a cursor object using the `cursor()` method of connection object.
3. The `execute()` methods run the SQL commands and return the result.
4. Use `cursor.fetchall()` or `fetchone()` or `fetchmany()` to read query result.
5. Use `commit()` to make the changes in database persistent, or use `rollback()` to revert the database changes.
6. Use `cursor.close()` and `connection.close()` method to close the cursor and PostgreSQL connection.

Source: `https://pynative.com/python-postgresql-tutorial/`

CREATE new PostgreSQL tables

```
import psycopg2
def createTable():
  conn = None
  try:
    conn = psycopg2.connect(database="mydb", user="myuser", password="mypass"
    cur = conn.cursor()
    cur.execute('''CREATE TABLE EMPLOYEE \
        (emp_num INT PRIMARY KEY NOT NULL, \
        emp_name VARCHAR(40) NOT NULL, \
        department VARCHAR(40) NOT NULL)''') # execute the CREATE TABLE statem
    conn.commit() # commit the changes to the database
    print ("Table created successfully")
    cur.close() # close the cursor
  except (Exception, psycopg2.DatabaseError) as error:
    print(error)
  finally
```

Executing INSERT statement from Python

```
import psycopg2
def insertRecord(num, name, dept):
  conn = None
  try:
    conn = psycopg2.connect(database="mydb", user="myuser", password="mypass"
    cur = conn.cursor()
    cur.execute("INSERT INTO EMPLOYEE (emp_num, emp_name, department) \
                 VALUES (%s, %s, %s)", (num, name, dept))
    conn.commit()
    print ("Total number of rows inserted :", cur.rowcount);
    cur.close()
  except (Exception, psycopg2.DatabaseError) as error:
    print(error)
  finally:
    if conn is not None: conn.close()
```

Executing DELETE statement from Python

```
import psycopg2
def deleteRecord(num):
  conn = None
  try:
    conn = psycopg2.connect(database="mydb", user="myuser", password="mypass"
    cur = conn.cursor()
    cur.execute("DELETE FROM EMPLOYEE WHERE emp_num = %s", (num,))
    conn.commit()
    print ("Total number of rows deleted :", cur.rowcount)
    cur.close()
  except (Exception, psycopg2.DatabaseError) as error:
    print(error)
  finally:
    if conn is not None: conn.close() # close the connection
deleteRecord(110) #function call
```

Executing UPDATE statement from Python

```
import psycopg2
def updateRecord(num, dept):
  conn = None
  try:
    conn = psycopg2.connect(database="mydb", user="myuser", password="mypass"
    cur = conn.cursor()
    cur.execute("UPDATE EMPLOYEE set department = %s where emp_num = %s", (dep
    conn.commit()
    print ("Total number of rows updated :", cur.rowcount)
    cur.close()
  except (Exception, psycopg2.DatabaseError) as error:
    print(error)
  finally:
    if conn is not None: conn.close()
updateRecord(110, "Finance") #function call
```

Executing SELECT statement from Python

```
import psycopg2
def selectAll():
    conn = None
    try:
        # connect to the PostgreSQL database
        conn = psycopg2.connect(database = "mydb", user = "myuser", \
            password = "mypass", host = "127.0.0.1", port = "5432")
        cur = conn.cursor()     # create a new cursor
        # execute the SELECT statement
        cur.execute("SELECT emp_num, emp_name, department FROM EMPLOYEE")
        rows = cur.fetchall()   # fetches all rows of the query result set
        for row in rows:
            print (print ("Employee ID = ", row[0], ", NAME = ", \
                row[1], ", DEPARTMENT = ", row[2]))
        cur.close()             # close the cursor
    except (Exception, psycopg2.DatabaseError) as error:
        print(error)
    finally:
        conn.close()            # close the connection
selectAll()                     # function call
```

Output:

```
Employee ID = 110, NAME = Bhaskar, DEPARTMENT = HR
Employee ID = 111, NAME = Ishaan, DEPARTMENT = FINANCE
Employee ID = 112, NAME = Jairaj, DEPARTMENT = TECHNOLOGY
Employee ID = 113, NAME = Ananya, DEPARTMENT = TECHNOLOGY
```

Python Frameworks for PostgresSQL

Web and Internet Development using Python

- Python offers several frameworks such as **bottle.py**, **Flask**, **CherryPy**, **Pyramid**, **Django** and **web2py** for web development.
- Python offers many choices for web development
  - Frameworks such as **Django** and **Pyramid**.
  - Advanced content management systems such as **Plone** and **django CMS**.
  - Micro-frameworks such as **Flask** and **Bottle**.
- Python's standard library supports many internet protocols
  - HTML and XML
  - E-mail processing
  - JSON
  - Support for FTP, IMAP, and other Internet protocols
  - Easy-to-use socket interface
- The package Index has more libraries
  - **Requests**, a powerful HTTP client library.
  - **Feedparser** for parsing RSS/Atom feeds.
  - **Beautiful Soup**, an HTML parser that can handle all sorts of HTML.
  - **Paramiko**, implementing the SSH2 protocol.
  - **Twisted Python**, a framework for asynchronous network programming

Flask Web Application Framework

- Flask is a lightweight WSGI (Web Server Gateway Interface) web application framework. It is designed to make getting started quick and easy, with the ability to scale up to complex applications.
- It began as a simple wrapper around Werkzeug (Werkzeug WSGI toolkit) and Jinja (Jinja template engine) and has since then become one of the most popular Python web application frameworks.
- Flask offers suggestions, but does not enforce any dependencies or project layouts. It is up to the developer to choose the tools and libraries they want to use.
- There are many extensions provided by the community that make adding new functionality easy.
- **Installing Flask using pip command**
  - `pip install -U Flask`

Source: `https://pypi.org/project/Flask/`

A Simple Example

```
from flask import Flask
app = Flask(__name__)

@app.route('/')
def hello_world():
  return "Hello World"

if __name__ == '__main__':
  app.run()
```

- Importing flask module in the project is mandatory. Our WSGI application is an object of `Flask` class.
- Flask constructor takes the name of current module (`__name__`) as argument.
- The `route()` function of the Flask class is a decorator, which tells the application which URL should call the associated function. `app.route(rule, options)`

A Simple Example (2)

`app.run(host, port, debug, options)`

- `host`: Hostname to listen on. Defaults to 127.0.0.1 (localhost). Set to `'0.0.0.0'` to have server available externally
- `port`: Defaults to 5000
- `debug`: Defaults to false. If set to true, provides a debug information
- `options`: To be forwarded to underlying Werkzeug server.

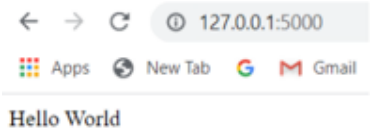
Executing the code:

- `python Hello.py`

Output: A message in Python shell:

- `* Running on http://127.0.0.1:5000/ (Press CTRL+C to quit)`

Open the above URL (127.0.0.1:5000) in the browser



Python: Flask

- Consider the table Candidate (in PostgreSQL) as shown below:

```
                    Table "public.candidate"
 Column |          Type          | Collation | Nullable | Default
--------+------------------------+-----------+----------+---------
 cno    | integer                |           | not null |
 name   | character varying(30)  |           |          |
 email  | character varying(30)  |           |          |
Indexes:
    "candidate_pkey" PRIMARY KEY, btree (cno)
```

- Code segment in Python:

```
from flask import Flask, render_template, request
import psycopg2

app = Flask(__name__, template_folder='templates')

# functions to be added here for different actions

if __name__ == '__main__': # Run the Flask app
    app.run(host='127.0.0.1', debug=True, port=5000)
```

Python: Flask (2)

- Source code for `index.html`:

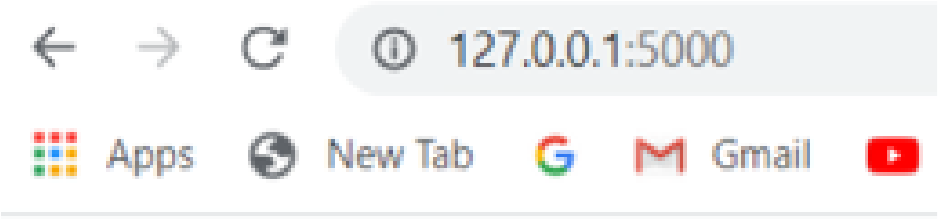
```
<!DOCTYPE html>
<html>
<head> <title>Candidate Email Database</title> </head>
<body>
  <h2>Candidate Email Database</h2>
  <a href="/add">Add Email</a><br><br>
  <a href="/viewall">View Email</a>
</body>
</html>
```

- Source code for rendering `index.html` and `add.html` pages:

```
@app.route("/")
def index():
    return render_template("index.html");
```

Python: Flask (3)

http://127.0.0.1:5000/



Python: Flask (4)

- Source code for `add.html` (in HTML):

```
<!DOCTYPE html>
<html>
<head> <title>Add Email</title> </head>
<body>
  <h2>Email Information</h2>
  <form action = "/savedetails" method="post">
    <table>
      <tr><td>CNO</td><td><input type="text" name="cno" required></td></tr>
      <tr><td>Name</td><td><input type="text" name="name" required></td></tr>
      <tr><td>Email</td><td><input type="text" name="email" required></td></tr
      <tr><td><input type="submit" value="Submit"></td></tr>
    </table>
  </form>
</body>
```

Python: Flask (5)

- saveDetails() function for add.html (in Python):

```
@app.route("/savedetails",methods = ["POST"])
def saveDetails():
  cno = request.form["cno"]
  name = request.form["name"]
  email = request.form["email"]
  conn = None
  try:
    conn = psycopg2.connect(database="mydb", user="myuser", password="mypass"
    cur = conn.cursor()
    cur.execute("INSERT INTO Candidate (cno, name, email) VALUES (%s, %s, %s)
    conn.commit()
    cur.close()
  except (Exception, psycopg2.DatabaseError) as error:
    return render_template("fail.html")
```

Python: Flask (6)

http://127.0.0.1:5000/add

CNO

Name

Email

Submit

http://127.0.0.1:5000/savedetails

Python: Flask (7)

- viewAll() function for viewall.html (in Python):

```
@app.route("/viewall")
def viewAll():
  conn = None
  try:
    conn = psycopg2.connect(database="mydb", user="myuser", password="mypass"
    cur = conn.cursor()
    cur.execute("SELECT cno, name, email FROM Candidate")
    results = cur.fetchall()
    cur.close()
  except (Exception, psycopg2.DatabaseError) as error:
    return render_template("fail.html")
  finally:
    if conn is not None: conn.close()
  return render_template("viewall.html",rows = results)
```

Python: Flask (8)

- Source code for `viewall.html` (in HTML):

```
<!DOCTYPE html>
<html>
<head> <title>Email List</title> </head>
<body>
  <h3>Email List</h3>
  <table border=5>
    <tr> <th>CNO</th><th>Name</th><th>Email</th> </tr>
    {% for row in rows %}
      <tr>
        <td>{{row[0]}}</td> <td>{{row[1]}}</td> <td>{{row[2]}}</td>
      </tr>
    {% endfor %}
  </table>
  <br><br>
```

Python: Flask (9)

http://127.0.0.1:5000/viewall

CNO	Name	Mobile
101	Ishaan	ishaan@mymail.com
102	Jairaj	Jairaj@mymail.com
103	Ananya	ananya@mymail.com
104	Barkha	barkha@myemail.com
105	Piyush	piyush@mymail.com

Module Summary

- Learnt how to access PostgreSQL database from Python
- Learnt to build Python Web Application with PostgresSQL and Flask

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